

Dynamic Setpoint Profiles / PLC_1 [CPU 1212C DC/DC/DC] / Program blocks

FB_TempRamp_Gen [FB1]

FB_TempRamp_Gen Properties							
General							
Name	FB_TempRamp_Gen	Number	1	Type	FB	Language	SCL
Numbering	Automatic						
Information							
Title		Author		Comment		Family	
Version	0.1	User-defined ID					

FB_TempRamp_Gen									
Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
▼ Input									
Start_Temp	Real	0.0	Non-retain	True	True	True	False		
Target_Temp	Real	0.0	Non-retain	True	True	True	False		
Ramp_Time_Min	Real	0.0	Non-retain	True	True	True	False		
Enable	Bool	false	Non-retain	True	True	True	False		
Reset	Bool	false	Non-retain	True	True	True	False		
dt_sec	Real	0.0	Non-retain	True	True	True	False		
▼ Output									
Output_Value	Real	0.0	Non-retain	True	True	True	False		
Ramp_Complete	Bool	false	Non-retain	True	True	True	False		
InOut									
▼ Static									
Elapsed_Time_sec	Real	0.0	Non-retain	True	True	True	False		
Ramp_Rate	Real	0.0	Non-retain	True	True	True	False		
Temp									
Constant									

```
0001 // 1. Calculate the rate of change in degrees per second
0002
0003 IF #Ramp_Time_Min > 0.0 THEN
0004     #Ramp_Rate := (#Target_Temp - #Start_Temp) / (#Ramp_Time_Min * 60);
0005 ELSE
0006     #Ramp_Rate := 0.0;
0007 END_IF;
0008
0009 // 2. Execution Logic
0010 IF #Reset THEN
0011     #Elapsed_Time_sec := 0.0;
0012     #Output_Value := #Start_Temp;
0013     #Ramp_Complete := FALSE;
0014 ELSIF #Enable AND NOT #Ramp_Complete THEN
0015     //// Accumulate elapsed time based on the execution cycle interval (dt)
0016     #Elapsed_Time_sec := #Elapsed_Time_sec + #dt_sec;
0017     // calculate the dynamic setpoint
0018     #Output_Value := #Start_Temp + (#Ramp_Rate * #Elapsed_Time_sec);
0019     // 3. Failsafe Clamping and Completion Check for both Heating and Cooling
0020     IF (#Target_Temp >= #Start_Temp AND #Output_Value >= #Target_Temp)
0021         OR (#Target_Temp < #Start_Temp AND #Output_Value <= #Target_Temp) OR
0022         (#Ramp_Time_Min <= 0.0) THEN
0023         #Output_Value := #Target_Temp;
0024         #Ramp_Complete := TRUE;
0025     END_IF;
0026 END_IF;
```

Symbol	Address	Type	Comment
#dt_sec		Real	
#Elapsed_Time_sec		Real	
#Enable		Bool	
#Output_Value		Real	
#Ramp_Complete		Bool	
#Ramp_Rate		Real	
#Ramp_Time_Min		Real	
#Reset		Bool	
#Start_Temp		Real	
#Target_Temp		Real	